TECHNICAL SUPPLEMENT

2028 Olympic Games Optimization

Prepared for SportsFanatics

This appendix documents the data and methods behind every figure in our presentation.

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FBLA · Data Analysis

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Each section is a self-contained reference. Appendices A–F give the full tables behind the figures shown on screen; the Methodology Summary states data sources, models, and assumptions for all three stakeholders.

Methodology Summary

Host City — Los Angeles (Appendix A)

Source: historical Olympic athlete-event records (1896–2016) joined to NOC region data. Reference Games: **Rio 2016 (Summer)**, the most recent complete Summer Games in the data. Each sport is mapped to its official LA28 venue zone (March 2026 venue plan); multi-venue sports are assigned to their primary sub-venue. We count **unique athletes** (by athlete ID) per zone, so no athlete is double-counted. Scope: the **31 returning LA28 sports**, recorded as 34 categories in the dataset (Rhythmic Gymnastics & Trampolining group under Gymnastics; Beach Volleyball under Volleyball). Total athletes analysed: **11,179**.

National Olympic Committees (Appendix B)

We add World Bank external data — **GDP per capita (NY.GDP.PCAP.CD)** and **population (SP.POP.TOTL)**, 2016 — and model Rio 2016 medal counts with a **Poisson regression** on log(population) and log(GDP per capita). Poisson is the correct family because medals are non-negative count data. Expected medals are estimated **out-of-fold** (5-fold CV, no leakage); the residual (actual − expected) is the Resource-Adjusted Performance Index (RAPI). Out-of-fold **R^2 = 0.80** over 182 countries. Each underperformer is matched to its nearest **overperforming** peer by Euclidean distance in log-population / log-GDP space.

Athletes (Appendices C & D)

Strategy clustering: KMeans (k=2) groups countries as specialist vs generalist by how concentrated their all-time Summer medals are across sports, using event-level medal counts and a concentration feature set (Herfindahl index + effective number of sports); countries with fewer than 20 medals are excluded as statistically unstable. **Peak age windows:** for each sport with at least 100 medal-winning athletes on record, we report the median medal-winning age and the interquartile range (middle 50%).

Consistency note. All medal-based figures use Rio 2016 as the reference Games and event-level medal counts (one medal per event, not per athlete) unless a section states an all-time scope. Every number in this document is computed directly from the datasets and notebooks in our repository.

Appendix A — Host-City Zone Load

Unique athletes per LA28 venue zone, Rio 2016 reference. Top three zones (Downtown/Exposition Park, Long Beach, Carson) carry **77%** of all 11,179 athletes. Load Index sets the heaviest zone = 100.

Zone	Athletes	Share	Sports	Load Index	Primary venue cluster
Downtown/Exposition Park	4,581	41.0%	12	100.0	LA Memorial Coliseum / Convention Center / Crypto.com Arena
Long Beach	2,455	22.0%	8	53.6	Long Beach Arena, Marine Stadium & Convention Center pool
Carson	1,526	13.7%	5	33.3	Dignity Health Sports Park (CSU Dominguez Hills)
Inglewood	1,223	10.9%	2	26.7	SoFi Stadium / Intuit Dome
Pasadena	609	5.4%	2	13.3	Rose Bowl & Rose Bowl Aquatics Center
Anaheim	283	2.5%	1	6.2	Honda Center
Arcadia	200	1.8%	1	4.4	Santa Anita Park
Pacific Palisades	120	1.1%	1	2.6	Riviera Country Club
Venice	110	1.0%	1	2.4	Venice Beach
San Fernando Valley	72	0.6%	1	1.6	Sepulveda Basin

Robustness — concentration is stable across recent Games

Recomputing zone shares for 2008 / 2012 / 2016 gives a Downtown share of **41% / 42% / 41%** and a top-3 share of **77% / 76% / 77%** — i.e. ~41% Downtown and ~77% top-3 in every Games. The allocation model does not depend on a single Games.

Sport-to-zone mapping (Rio 2016 athletes per sport)

Sport	Zone	Ath.
Volleyball	Anaheim	283
Equestrianism	Arcadia	200
Cycling	Carson	513
Hockey	Carson	390
Rugby Sevens	Carson	299
Tennis	Carson	196
Archery	Carson	128
Athletics	Downtown/Expo	2,269
Judo	Downtown/Expo	389
Wrestling	Downtown/Expo	346
Boxing	Downtown/Expo	283
Weightlifting	Downtown/Expo	255
Fencing	Downtown/Expo	245
Gymnastics	Downtown/Expo	196
Table Tennis	Downtown/Expo	172
Badminton	Downtown/Expo	172

Sport	Zone	Ath.
Rhythmic Gymnastics	Downtown/Expo	96
Trampolining	Downtown/Expo	32
Swimming	Inglewood	942
Basketball	Inglewood	281
Rowing	Long Beach	546
Shooting	Long Beach	390
Sailing	Long Beach	380
Handball	Long Beach	353
Canoeing	Long Beach	330
Water Polo	Long Beach	258
Synchronized Swimming	Long Beach	102
Beach Volleyball	Long Beach	96
Golf	Pacific Palisades	120
Football	Pasadena	473
Diving	Pasadena	136
Modern Pentathlon	San Fernando Valley	72

Sport	Zone	Ath.
Taekwondo	Downtown/Expo	126

Sport	Zone	Ath.
Triathlon	Venice	110

Appendix B — NOC Over / Underperformers

Poisson resource-adjusted model, Rio 2016. Out-of-fold R^2 = **0.80**, MAE = 3.13 medals, n = 182 countries. RAPI = actual − expected medals. GDP = GDP per capita (USD, 2016); Pop in millions.

B.1 — Overperformers (all 44, by RAPI)

Country	Act.	Exp.	RAPI	GDP/cap	Pop (M)
Russia	56	15.3	+40.7	8,663	145.0
UK	67	30.4	+36.6	41,258	65.6
Azerbaijan	18	1.5	+16.5	3,881	9.8
Kazakhstan	18	3.6	+14.4	7,476	18.4
New Zealand	18	6.0	+12.0	40,114	4.7
Hungary	15	3.7	+11.3	13,216	9.8
Kenya	13	2.1	+10.8	1,554	48.1
Uzbekistan	13	2.4	+10.6	2,753	31.3
Jamaica	11	0.8	+10.2	5,302	2.8
Australia	29	20.3	+8.7	50,084	24.2
Ukraine	11	2.4	+8.7	2,158	45.6
France	42	33.7	+8.3	37,024	66.7
Cuba	11	2.8	+8.2	8,111	11.3
Croatia	10	2.0	+8.0	12,820	4.1
Denmark	15	7.5	+7.5	54,501	5.7
Belarus	9	1.9	+7.1	5,040	9.5
Serbia	8	1.8	+6.2	5,982	7.1
Ethiopia	8	1.8	+6.2	642	106.7
Georgia	7	0.8	+6.2	4,143	3.7
Germany	42	36.4	+5.6	42,949	82.3
Czech Republic	10	5.6	+4.4	18,754	10.6
Armenia	4	0.7	+3.4	3,524	3.0
Netherlands	19	15.7	+3.3	46,809	17.0
Lithuania	4	1.8	+2.2	14,934	2.9
Italy	28	25.9	+2.1	31,392	60.1
Colombia	8	6.1	+1.9	5,960	47.4
Slovenia	4	2.1	+1.9	21,448	2.1
Poland	11	9.4	+1.6	12,464	38.0
Tunisia	3	1.6	+1.4	3,848	11.5
Mongolia	2	0.6	+1.4	3,620	3.1
Canada	22	20.9	+1.1	42,314	36.1
Bahamas	2	0.9	+1.1	30,617	0.4
Greece	6	5.0	+1.1	17,919	10.8
Bulgaria	3	2.2	+0.8	7,822	6.9
Burundi	1	0.2	+0.8	233	11.2
Grenada	1	0.2	+0.8	9,221	0.1
Slovakia	4	3.2	+0.8	16,636	5.4
Tajikistan	1	0.4	+0.6	789	8.9
Fiji	1	0.4	+0.6	5,364	0.9
Niger	1	0.5	+0.5	503	20.7
Kosovo	1	0.5	+0.5	3,759	1.8
China	70	69.6	+0.4	8,255	1387.8
Ivory Coast	2	1.7	+0.3	1,863	26.0
Bahrain	2	1.9	+0.1	23,800	1.4

B.2 — Underperformers (expected ≥ 1 medal; 66 of 138 total, by RAPI)

Country	Act.	Exp.	RAPI	GDP/cap	Pop (M)
India	2	21.6	-19.6	1,708	1343.9
USA	121	137.9	-16.9	57,638	324.4
Saudi Arabia	0	13.5	-13.5	22,268	31.0
Japan	41	54.4	-13.4	39,375	127.1
Mexico	5	15.6	-10.6	9,098	122.3
Austria	1	10.1	-9.1	45,061	8.7
Indonesia	3	11.8	-8.8	3,521	264.6
Algeria	2	4.2	-2.2	4,425	40.9
Uruguay	0	2.2	-2.2	17,010	3.4
Guatemala	0	2.2	-2.2	4,060	16.3
Lebanon	0	1.9	-1.9	8,089	6.3
Angola	0	1.9	-1.9	2,052	29.2
Libya	0	1.9	-1.9	7,526	6.6
Ghana	0	1.8	-1.8	1,912	29.4

Country	Act.	Exp.	RAPI	GDP/cap	Pop (M)
United Arab Emirates	1	9.6	-8.6	42,721	8.9
Switzerland	7	15.4	-8.4	82,153	8.4
Ireland	2	8.8	-6.8	64,130	4.8
Israel	2	8.3	-6.3	37,672	8.5
Finland	1	7.2	-6.2	43,451	5.5
Norway	4	10.2	-6.2	70,867	5.2
Chile	0	5.9	-5.9	13,650	18.3
Argentina	4	9.7	-5.7	12,700	43.9
Nigeria	1	6.7	-5.7	2,070	195.4
Belgium	6	11.1	-5.0	41,855	11.3
Bangladesh	0	5.0	-5.0	1,649	160.8
Philippines	1	6.0	-5.0	2,985	106.7
Peru	0	4.9	-4.9	6,217	30.9
Portugal	1	5.8	-4.8	19,980	10.3
Pakistan	0	4.6	-4.6	1,425	220.1
Iraq	0	4.0	-4.0	4,334	38.5
Qatar	1	4.7	-3.7	61,254	2.5
Brazil	19	22.7	-3.7	8,836	203.2
Luxembourg	0	3.4	-3.4	106,899	0.6
Egypt	3	6.2	-3.2	3,271	101.6
Oman	0	3.0	-3.0	17,110	4.4
Ecuador	0	3.0	-3.0	5,918	16.5
Vietnam	2	4.5	-2.5	2,735	94.0
Sri Lanka	0	2.5	-2.5	4,149	21.2
Costa Rica	0	2.4	-2.4	12,091	4.9
Panama	0	2.2	-2.2	14,832	4.0

Country	Act.	Exp.	RAPI	GDP/cap	Pop (M)
Cyprus	0	1.7	-1.7	24,280	1.2
Morocco	1	2.7	-1.7	3,141	35.0
Myanmar	0	1.6	-1.6	1,229	51.5
Turkmenistan	0	1.6	-1.6	5,687	6.4
Iceland	0	1.6	-1.6	62,854	0.3
Paraguay	0	1.5	-1.5	5,775	6.2
Tanzania	0	1.5	-1.5	953	53.8
Dominican Republic	1	2.5	-1.5	7,160	10.6
Sudan	0	1.3	-1.3	1,033	41.3
Latvia	0	1.3	-1.3	13,839	2.0
Cameroon	0	1.3	-1.3	1,442	23.5
Boliva	0	1.3	-1.3	3,036	11.2
Democratic Republic of the Congo	0	1.2	-1.2	460	84.0
Malaysia	5	6.2	-1.2	9,477	31.8
El Salvador	0	1.1	-1.1	3,901	6.2
Cambodia	0	1.1	-1.1	1,675	15.9
Uganda	0	1.1	-1.1	753	38.8
South Korea	21	22.1	-1.1	30,832	51.2
Romania	4	5.0	-1.0	9,405	19.7
Thailand	6	7.0	-1.0	5,834	70.9
Sweden	11	11.9	-0.9	51,704	9.9
Jordan	1	1.6	-0.6	3,987	10.0
Spain	17	17.5	-0.5	26,756	46.5
Iran	8	8.5	-0.5	5,711	83.8
Estonia	1	1.4	-0.4	18,666	1.3
Venezuela	3	3.2	-0.2	3,670	30.8

B.3 — Peer Benchmark: nearest overperforming peer

For each underperformer (expected ≥ 1 medal), the nearest **overperforming** country by resource distance (Euclidean in log-population / log-GDP space). Sorted by distance — the closest, most actionable comparisons are first.

Featured peer chain — Jordan to Azerbaijan. Jordan (RAPI -0.6) is the tightest match in the entire table: its nearest overperformer is **Azerbaijan** (distance 0.037; near-identical population and GDP per capita) which converts the same resource base into 18 medals by concentrating on a few combat-sport disciplines. **Belarus** is also a comparable overperforming peer (distance 0.241, 9 medals) — either supports the "concentrate investment on winnable disciplines" recommendation.

Country	RAPI	Nearest peer	P.RAPI	P.med	Dist.
Jordan	-0.6	Azerbaijan	+16.5	18	0.037
Libya	-1.9	Bulgaria	+0.8	3	0.055
Portugal	-4.8	Czech Republic	+4.4	10	0.067
Lebanon	-1.9	Bulgaria	+0.8	3	0.093
Turkmenistan	-1.6	Serbia	+6.2	8	0.116
Ghana	-1.8	Ivory Coast	+0.3	2	0.125
Paraguay	-1.5	Serbia	+6.2	8	0.127
Dominican Republic	-1.5	Cuba	+8.2	11	0.141
Cyprus	-1.7	Bahrain	+0.1	2	0.142
Argentina	-5.7	Poland	+1.6	11	0.146
Panama	-2.2	Croatia	+8.0	10	0.147
Angola	-1.9	Ivory Coast	+0.3	2	0.151
South Korea	-1.1	Italy	+2.1	28	0.161
Morocco	-1.7	Uzbekistan	+10.6	13	0.173
Finland	-6.2	New Zealand	+12.0	18	0.173
Mexico	-10.6	Russia	+40.7	56	0.178
Costa Rica	-2.4	Croatia	+8.0	10	0.180
Uruguay	-2.2	Lithuania	+2.2	4	0.207
Oman	-3.0	Slovakia	+0.8	4	0.214
Boliva	-1.3	Tunisia	+1.4	3	0.239
Romania	-1.0	Kazakhstan	+14.4	18	0.240
Myanmar	-1.6	Kenya	+10.8	13	0.244
Ireland	-6.8	Denmark	+7.5	15	0.246
Estonia	-0.4	Bahrain	+0.1	2	0.255
Ecuador	-3.0	Kazakhstan	+14.4	18	0.257

Country	RAPI	Nearest peer	P.RAPI	P.med	Dist.
Iraq	-4.0	Colombia	+1.9	8	0.381
Latvia	-1.3	Lithuania	+2.2	4	0.392
Thailand	-1.0	Colombia	+1.9	8	0.402
Democratic Republic of the Congo	-1.2	Ethiopia	+6.2	8	0.411
Belgium	-5.0	Netherlands	+3.3	19	0.422
Peru	-4.9	Colombia	+1.9	8	0.432
Sudan	-1.3	Kenya	+10.8	13	0.436
Japan	-13.4	Germany	+5.6	42	0.442
El Salvador	-1.1	Serbia	+6.2	8	0.447
Austria	-9.1	Denmark	+7.5	15	0.463
Tanzania	-1.5	Kenya	+10.8	13	0.502
Cambodia	-1.1	Ivory Coast	+0.3	2	0.506
United Arab Emirates	-8.6	Denmark	+7.5	15	0.507
Israel	-6.3	Denmark	+7.5	15	0.544
Sweden	-0.9	Netherlands	+3.3	19	0.549
Switzerland	-8.4	Denmark	+7.5	15	0.559
Sri Lanka	-2.5	Uzbekistan	+10.6	13	0.565
Iran	-0.5	Colombia	+1.9	8	0.571
Chile	-5.9	Greece	+1.1	6	0.594
Saudi Arabia	-13.5	Poland	+1.6	11	0.615
Iceland	-1.6	Bahamas	+1.1	2	0.734
Uganda	-1.1	Niger	+0.5	1	0.747
Vietnam	-2.5	Ukraine	+8.7	11	0.761
Qatar	-3.7	New Zealand	+12.0	18	0.770
Egypt	-3.2	Ukraine	+8.7	11	0.903

Country	RAPI	Nearest peer	P.RAPI	P.med	Dist.
Cameroon	-1.3	Ivory Coast	+0.3	2	0.276
Norway	-6.2	Denmark	+7.5	15	0.278
Venezuela	-0.2	Uzbekistan	+10.6	13	0.288
Spain	-0.5	Italy	+2.1	28	0.303
Malaysia	-1.2	Poland	+1.6	11	0.327
Algeria	-2.2	Colombia	+1.9	8	0.333
Brazil	-3.7	Russia	+40.7	56	0.338
Guatemala	-2.2	Tunisia	+1.4	3	0.349

Country	RAPI	Nearest peer	P.RAPI	P.med	Dist.
Philippines	-5.0	Ukraine	+8.7	11	0.910
Bangladesh	-5.0	Ethiopia	+6.2	8	1.029
Pakistan	-4.6	Ethiopia	+6.2	8	1.077
Indonesia	-8.8	Russia	+40.7	56	1.083
Luxembourg	-3.4	Bahamas	+1.1	2	1.314
Nigeria	-5.7	Ethiopia	+6.2	8	1.318
USA	-16.9	Germany	+5.6	42	1.402
India	-19.6	China	+0.4	70	1.576

Appendix C — Country Strategy Classification

KMeans (k=2) on all-time Summer event-level medals, classifying countries as **Specialist** (medals concentrated in 1–2 sports) or **Generalist** (spread across many). 62 countries meet the ≥ 20-medal threshold (22 specialist, 40 generalist). **Top-sport %** = share of a country's medals in its single best sport; **Eff. sports** = effective number of sports (1 / Herfindahl index) — lower means more specialised.

Specialists (by effective number of sports, most concentrated first)

Country	Med.	Top sport	Top %	Eff. sp.
Ethiopia	53	Athletics	100%	1.0
Jamaica	76	Athletics	99%	1.0
Kenya	100	Athletics	93%	1.1
Morocco	23	Athletics	83%	1.4
Trinidad	21	Athletics	81%	1.5
Indonesia	30	Badminton	63%	1.9
Iran	68	Wrestling	63%	2.1
Turkey	95	Wrestling	66%	2.1
Slovakia	28	Canoeing	64%	2.2
Georgia	32	Wrestling	56%	2.5
Thailand	30	Boxing	47%	2.6
Nigeria	24	Athletics	54%	2.7
Azerbaijan	44	Wrestling	52%	3.1
Uzbekistan	33	Boxing	42%	3.2
Ireland	33	Boxing	48%	3.4
Mongolia	26	Wrestling	35%	3.4
Taiwan	24	Taekwondo	33%	4.0
Egypt	27	Weightlifting	41%	4.1
Portugal	24	Athletics	42%	4.3
Finland	310	Athletics	37%	4.3
India	28	Hockey	39%	4.5
Kazakhstan	70	Boxing	31%	4.6

Generalists (most diversified first)

Country	Med.	Top sport	Top %	Eff. sp.
South Africa	88	Athletics	32%	4.9
Estonia	34	Wrestling	32%	5.0
Australia	518	Swimming	39%	5.1
Cuba	214	Boxing	34%	5.1
North Korea	54	Weightlifting	32%	5.3
Slovenia	23	Judo	22%	5.9
USA	2544	Athletics	32%	6.1
Bulgaria	217	Wrestling	32%	6.2
Colombia	28	Cycling	21%	6.5
Argentina	74	Boxing	32%	6.7
Japan	437	Gymnastics	22%	6.7
New Zealand	118	Athletics	20%	6.9
Mexico	67	Diving	21%	7.4
Norway	155	Shooting	21%	8.1
Romania	303	Gymnastics	23%	8.3
Switzerland	209	Gymnastics	23%	8.6
Lithuania	25	Athletics	20%	8.6
Netherlands	297	Swimming	20%	9.4
Canada	305	Athletics	20%	9.4
Czech Republic	207	Athletics	18%	9.5
Brazil	128	Judo	17%	9.7
Poland	290	Athletics	20%	9.8
Sweden	513	Athletics	18%	10.0
UK	892	Athletics	24%	10.1
Croatia	33	Athletics	15%	10.2
South Korea	263	Judo	16%	10.2
Russia	1577	Athletics	18%	10.7
Italy	608	Fencing	21%	11.0
Germany	1389	Athletics	19%	11.2
Serbia	107	Wrestling	16%	11.4
Belgium	166	Cycling	16%	12.2
Austria	108	Canoeing	13%	12.4
Ukraine	126	Athletics	14%	12.5
Denmark	210	Sailing	14%	12.5

Country	Med.	Top sport	Top %	Eff. sp.
Greece	151	Athletics	24%	8.8
Belarus	84	Athletics	20%	8.9
Hungary	504	Fencing	18%	9.0

Country	Med.	Top sport	Top %	Eff. sp.
China	545	Diving	13%	12.7
France	770	Fencing	16%	13.1
Spain	148	Sailing	13%	14.7

Footnote — low-medal countries. Countries with fewer than 20 event-level medals are excluded because share-based concentration is unstable at low counts. By inspection these are typically clear specialists: e.g. **Pakistan** (10 medals, 80% in Hockey) is intuitively a specialist but falls below threshold, so it is reported here rather than given an unstable cluster label. Labels reflect degree of concentration, not which sport — a country specialised in an unusual sport (e.g. India in field hockey) is correctly a specialist.

Appendix D — Peak Age Windows per Sport

Median medal-winning age and interquartile range (middle 50%) for every Summer sport with at least **100** medal-winning athletes on record, sorted youngest to oldest. Q1 / Q3 are the 25th / 75th age percentiles; IQR = Q3 − Q1. Discontinued events (Art Competitions, 15-a-side Rugby) are excluded. Sample size n = medal-winning athletes.

Sport	Med.	Q1	Q3	IQR	n
Rhythmic Gymnastics	18	17	20	3.2	132
Swimming	21	18	23	5.0	3,027
Diving	22	19	25	6.0	423
Taekwondo	23	21	26	5.0	144
Synchronized Swimming	23	22	26	4.0	215
Boxing	23	21	25	4.0	931
Football	23	22	27	5.0	1,531
Gymnastics	23	20	27	7.0	2,187
Athletics	24	22	27	5.0	3,945
Cycling	24	21	27	6.0	1,235
Table Tennis	24	22	27	5.0	168
Rowing	25	22	28	6.0	2,830
Weightlifting	25	22	28	6.0	636
Hockey	25	23	28	5.0	1,497
Judo	25	23	27	4.0	547
Volleyball	25	23	28	5.0	969
Basketball	25	22	28	6.0	1,078
Water Polo	25	23	29	6.0	1,030
Badminton	25	23	27	4.2	168
Tennis	26	23	30	7.0	328
Softball	26	24	29	5.0	179
Wrestling	26	23	28	5.0	1,265
Handball	26	23	29	6.0	1,060
Canoeing	26	23	29	6.0	1,165
Baseball	26	23	29	6.0	336
Modern Pentathlon	27	25	29	4.0	186
Archery	27	22	38	16.0	319
Fencing	28	24	32	8.0	1,720
Sailing	30	26	36	10.0	1,175
Shooting	32	27	39	12.0	1,167
Equestrianism	34	29	41	12.0	955

Early-peak / tight-window sports (e.g. Rhythmic Gymnastics, median 18) reward physical peak; late-peak sports (Shooting 32, Equestrianism 34) reward accumulated experience — development pipelines should be sport-specific.

Appendix E — New LA28 Sports (scope addendum)

Five sports are added for LA28. **Four — cricket, flag football, lacrosse, squash — are absent from the historical data**, so they sit outside the quantitative zone model; the estimates below are forward-looking, anchored to comparable existing sports (Rio 2016 athlete counts). **Baseball and softball** were Olympic sports through Beijing 2008 and do appear in the data, so their figures are observed historical counts, not estimates.

New sport	LA28 venue / zone (per LA28 plan)	Comparable sport	Est. athlete load
Cricket (T20)	Fairgrounds — Pomona	Field Hockey (390)	≈ 390 (mid-size team sport)
Baseball	Stadium venue (returning sport)	Observed, Beijing 2008	191 (historical)
Softball	Stadium venue (returning sport)	Observed, Beijing 2008	120 (historical)
Lacrosse (Sixes)	Field venue — Carson cluster	Rugby Sevens (299)	≈ 299 (small-side team sport)
Flag Football	Stadium venue	Rugby Sevens / Basketball	≈ 281 (small-roster teams)
Squash	Temporary glass court	Table Tennis / Badminton (172)	≈ 172 (individual racket)

How this affects the core analysis. The added sports raise total athlete load but are spread across outlying / new zones (Pomona, stadium and field venues) rather than the three high-load zones, so they do not change the central finding: load remains concentrated in Downtown/Exposition Park, Long Beach, and Carson. Venue placements follow LA28 announcements and are noted for planning context only; they are outside our quantitative model.

Appendix F — Data Sources & Reproducibility

Dataset	Detail
Olympic athlete events	athlete_events.csv — 271,116 athlete-event rows, 1896–2016 (Kaggle / Sports Reference)
NOC regions	noc_regions.csv — National Olympic Committee to country mapping
World Bank — GDP per capita	Indicator NY.GDP.PCAP.CD (current US$), 2016
World Bank — Population	Indicator SP.POP.TOTL (total population), 2016
LA28 venue plan	Official LA28 venue-to-sport assignments (March 2026 map)

Tools & libraries

Python · pandas · NumPy · scikit-learn (PoissonRegressor, KMeans, KFold cross-validation) · SciPy · Matplotlib · Jupyter. Models: Poisson regression (NOC medal model) and KMeans clustering (country strategy).

Reproducibility

Every figure in this supplement and in the presentation traces to the notebooks in our GitHub repository: LA28_zone_load_analysis (Appendix A), gdpRegression (Appendix B), olympics_country_specialization_clustering (Appendix C), and olympics_analysis (Appendix D). A single script (build_supplement_data.py) regenerates every number in this document from the raw datasets, so SportsFanatics can audit or extend the analysis.